

此处是数字逻辑基础（杨旭老师）2025-spring final 的回忆版试卷。

介绍：考试会给你一定的小情景，就是两个人在一起对话，然后你作为回答的一方去回答问题，相应的完成考试。

经验：不要过度在乎小字的部分，其实没有信息，只读黑体即可，不然容易做不完题目，备考上建议 PPT 上的题目与作业题全部做一遍，基本上就可以。

关于考试：做题速度快些为好，连线很痛苦，因为会一团乱麻。

备注：此处回忆版作为免费开源公开，请资料售卖者不要下载并售卖，虽然我知道说了也没什么用。

The fundamental of Digital logic

Part 1

Given number is 51.125

1. Convert to binary
2. Conver to 5421BCD code
3. 5421BCD code addition, given two number and get the sum of them.

Part 2 basic logic

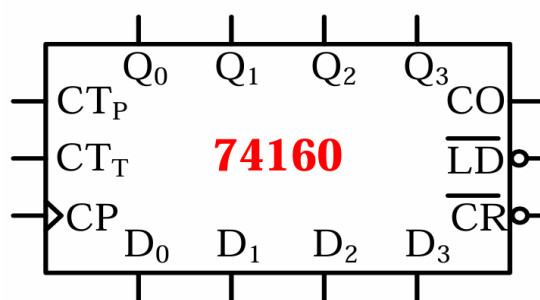
1. Based on the truth table , write the standard SOP and POS.
2. Use Boolean algebra to simplify the logical expression
3. Use k-map to simplify the logical expression.

Part 3 combinational logic

1. Use the basic logic gate to compose the circuit, including logic expression abstract ,simplify and final implementation. (but I found the result is $Z = D$, which means I don't need any gate then get the result? Strange.)
2. Use the 74LS151 to implement the function.
3. Use the 74LS138 to implement the function.
4. (a little difficult I think) use the 4-bits full adder to implement the function, which asks you to convert excess-3 code to 5421BCD code.(yes 5421BCD again lol)

Part 4 sequential logic

1. Write the state function of JK flip-flop.
2. Write the state function of T flip-flop.
3. Use the D flip-flop and basic logic gate to implement 7-arry counter. (final part of the slices of PPT-7), you should draw the D and other lines.
4. Use the 74610 (decimal counter) and basic logic gate to implement a function (which is similar with your lab2, but you can't use the decoder.), you should implement the 7-art counter first and then make the 3 lights on in different condition.



CP顺序	红	黄	绿
0	0	0	0 ←
1	1	0	0
2	0	1	0
3	0	0	1
4	1	1	1
5	0	0	1
6	0	1	0